Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, turning owners' growth goals into site-selection analyses and revenue targets; identified a $1.5M expansion opportunity.
- Built a revenue driver tree (traffic × capture rate × ticket price, with a traffic-to-capture elasticity) and translated client profit goals into break-even revenue and volume targets.
- Estimated the incremental revenue of new locations with Double ML and causal forests (econml, XGBoost nuisance models) on a 12-year county panel, stating identification assumptions and checking robustness with difference-in-differences and a synthetic-data recovery test.
- Back-tested demand predictions out of sample against real operating sites, measuring bias, error, rank correlation, and 90% interval coverage before using the model for planning.
- Built low/base/high scenarios and Monte Carlo risk models (probability of loss, VaR, CVaR) that carry the causal estimates' 95% confidence intervals, and explained sensitivity and uncertainty to owners in written reports.
- Engineered reproducible Python pipelines with data-quality checks and 150+ unit tests, cutting analysis turnaround by 91%, and documented each method in reusable walkthroughs.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, engineering, and behavioral science leadership, setting OKRs and measurement frameworks that tied analytics work to business goals.
- Standardized cross-team A/B testing and causal inference frameworks for direct-mail campaigns so every team measured impact the same way, applying CUPED to reduce metric variance by 20%.
- Built uplift models on millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible records to target incremental (causal) response rather than predicted response, cutting ad spend by $10K per campaign.
- Analyzed a monthly retention survey, linking member responses to retention behavior with driver and segment analysis in recurring monthly reporting.
- Owned the monthly model monitoring report on drift, performance, and media mix forecast-vs-actual accuracy for up to 10 production models, with scheduled Databricks jobs refreshing the metrics.
- Re-architected a production churn prediction platform in Python, SQL, Spark, Snowflake, and Databricks, reducing runtime by 48% and memory use by 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics, evaluation frameworks, and drift monitoring for production search and pricing models, using offline validation and online experiments, and mentored the team on ML practices.
- Built SQL training tables that broke bulk orders into item-level shipping costs, then partnered with engineering to ship a regression-based pricing API in C# on Azure, scaling to 10,000+ daily requests.
- Reduced customer dissatisfaction with search by 12% with product and engineering partners through custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built and validated XGBoost classification models on millions of consumer records with hundreds of external attributes, driving a 3% lift in response rates, and explained model drivers to stakeholders with SHAP.
- Automated reporting pipelines and Power BI dashboards with Python, SQL, and Snowflake, removing 60 minutes of manual work daily.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas, focusing campaign resources on high-opportunity regions.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in optimization, functional analysis, and operator theory, and taught mathematics and statistics to classes of 75-200 students.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show when the LTV = ARPU / churn identity breaks down (tenure-dependent churn), and estimated lifetime value from empirical survival curves instead.

Mobile Game Retention A/B Test github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Ran a full experiment workflow on 1- and 7-day player retention: hypotheses, outlier handling, sample sizing, significance testing, and practical significance for the launch call.

Medicare Claims Intelligence github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a tested dbt warehouse (staging to marts) over 790K CMS claims, cut forecast error 41% vs. a seasonal benchmark in rolling backtests (ARIMA, ETS, Prophet), and built a SHAP-explained model that finds 46% more high-cost members.

CaLaun — *AI Calculus Tutor* github.com/calaun-learning/calaun

- Built and deployed an LLM tutor grounded in SymPy-verified steps, serving 400+ weekly users with 90+ automated tests.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Trees & Driver Decomposition, Metric Definitions, Retention & Cohort Analysis, Survival Analysis, LTV / ARPU, Sensitivity & Scenario Analysis, Target Setting, Forecast Validation (Backtesting)

Causal Inference & Experimentation: A/B Testing, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, CATE / Uplift Modeling, Robustness Checks, Monte Carlo Simulation

Modeling: Regression, Classification, XGBoost, Time Series Forecasting (ARIMA, ETS, Prophet), Media Mix Modeling, Graph Mining (NetworkX), SHAP, Model Monitoring & Drift

Data & Tools: SQL, Python (pandas, NumPy, statsmodels, scikit-learn, econml), dbt, Snowflake, Spark, Databricks, SQL Server, PostgreSQL, AWS, Azure, Git, Unit Testing, Power BI, Tableau, Excel, AI-Assisted Development (Claude Code)